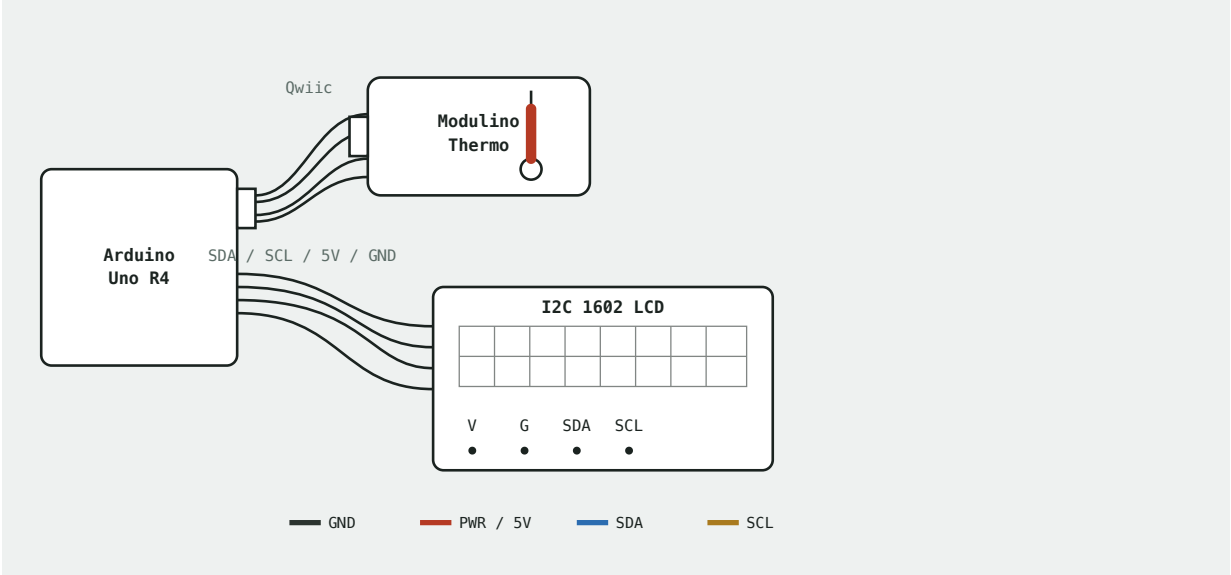
I2C LCD weather display

Modulino Thermo reports temperature and humidity; a separate I2C 1602 character LCD shows them. The Uno R4 has two I2C buses: the Modulino rides the Qwiic connector while the LCD runs from the SDA/SCL header pins at 5V.

Board · Arduino Uno R4 **Modules** · Modulino Thermo, I2C 1602 LCD
New concepts · two I2C buses, `millis()` timing



The Thermo plugs into the Qwiic connector (`Wire1`, 3.3V). The LCD is wired separately to the header: GND, **5V**, SDA, SCL (`Wire`, same bus as A4/A5). Different buses, so the two devices never get in each other's way.

PIN & CONNECTION REFERENCE

Component	Connects to
Modulino Thermo	Qwiic/I2C bus (Arduino's Qwiic connector)
I2C 1602 LCD — VCC	Arduino 5V
I2C 1602 LCD — GND	Arduino GND
I2C 1602 LCD — SDA	Arduino SDA (header pin next to AREF)
I2C 1602 LCD — SCL	Arduino SCL (header pin next to SDA)

Gotcha: the LCD needs **5V** — at 3.3V it stays blank. A completely blank screen is otherwise almost always the contrast trim-pot (on the back of most I2C backpacks) or the wrong address — common ones are `0x27` and `0x3F`. Run an "I2C Scanner" sketch to find the real one.

REQUIRED LIBRARIES

- `Arduino_Modulino` — pulls in the sensor-vendor dependencies automatically
- `LiquidCrystal_I2C` — Library Manager

CODE REFERENCE

Output — write to the I2C LCD

No pin list to wire up — just the address, columns and rows. The declaration goes at the top of the sketch, `init`/`backlight` in `setup()`, and `setCursor`/`print` wherever you update the screen (e.g. in `loop()`).

```
// --- Variables (top of sketch) ---
LiquidCrystal_I2C lcd(0x27, 16, 2);

// --- Inside setup() ---
lcd.init();
lcd.backlight();

// --- Inside loop() ---
lcd.setCursor(0, 0);
lcd.print("Hello!");
```

Timing — update without `delay()`

Remember the last time, then check the elapsed time. The two variables go at the top of the sketch; the `if` goes in `loop()`.

```
// --- Variables (top of sketch) ---
const unsigned long INTERVAL_MS = 2000;
unsigned long lastReadTime = 0;

// --- Inside loop() ---
if (millis() - lastReadTime >=
INTERVAL_MS) {
  lastReadTime = millis();
  showReading();
}
```

TRY IT YOURSELF

- Add a Fahrenheit readout on a rotating third line (alternate what `showReading()` prints every few seconds).
- Print a small text "bar" of asterisks on line 2 whose length scales with the humidity.